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Preparation of electroless Ni-P alloy coating with medium temperature and low phosphorus content

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Low phosphorus electroless Ni-P alloy coating has many advantages over medium and high phosphorus electroless Ni-P alloy coatings because of its special microstructure and composition, which can meet more special performance requirements. In addition, reducing the plating temperature can reduce energy consumption. The preparation process of electroless Ni-P plating at medium temperature and low-P was studied by single factor and orthogonal experiment. The results show that electroless Ni-P alloy coatings with 4–6% P can be obtained at 60–80°C and pH=5–6. The hardness, deposition rate and corrosion resistance of the coating were also evaluated.

Keywords: Medium temperature; electroless plating; low phosphorus; Ni-P alloy.

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1. Introduction

The electroless nickel plating process is simple to operate and the coating has excellent coating performance, good corrosion resistance and wear resistance.[1,2] As a protective, decorative and functional coating, electroless plating is widely used in machinery, electronics, plastics, molds, petrochemicals, textiles, aerospace and

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other industries.[3,4] Therefore, electroless plating is a surface treatment method with broad prospects for applications. Traditional electroless nickel plating is carried out at a relatively high-temperature (90–95°C). It has a fast deposition rate.[5,6] However, high-temperature requires higher plating bath and heating, consuming more energy, and resulting in poor stability and short-service life of plating solution, difficult operation and poor working condition.[7–9]

Every 10 K reduction of temperature in electroless plating operation not only reduces the cost by 5–10%, but also reduces the evaporation and acid mist of plating solution, which can improve the working environment and is easy to realize large workpiece plating.[10,11] Therefore, electroless Ni-P plating at medium temperature is of great significance.[12] In addition, P content in electroless Ni plating is an important factor affecting the crystalline orientation of the coating. Low and medium P electroless Ni plating is a mixture of amorphous and microcrystalline Ni phases, while high P electroless Ni plating is amorphous.[13,14] With increasing P content, the density of electroless Ni-P plating layer decreases gradually, and its weldability becomes worse.

In this work, the preparation process of electroless Ni-P plating at medium temperature and low P has been studied by single factor experiment and orthogonal experiment. The aim is to find a medium temperature and low P process formulation, and optimize its hardness, deposition speed and corrosion resistance.

2. Experimental Methods

The substrate specimen used in the experiment was Q235 cold-rolled steel sheet with a size of $40 \times 25 \times 3$ mm^3. Prior to electroless plating, the surface of the specimen was properly treated with polishing and chemical cleaning.[15] In order to optimize the plating parameters, the amount of $NaH_2PO_2{\cdot}H_2O$ and CH_3COONa added, plating temperature and solution pH were studied as the factors in a four-factor three-level orthogonal test. The amounts of $NaH_2PO_2{\cdot}H_2O$ added were 30, 35 and 40 g/L, respectively. The amounts of CH_3COONa added were 15, 20 and 25 g/L. The plating temperature was 60°C, 70°C and 80°C. The pH was 5, 5.5 and 6. The variables that were assessed included the deposition rate of the plating layer and P content.

The hardness value of the coating was measured by the MHV-1000Z-type microhardness tester. The morphology and elemental composition of coatings was observed on a Scanning Electron Microscope with an Energy Dispersive X-ray spectrometer. The deposition rate was measured by the mass-gain method as follows:

$$v = (W_2 - W_1)/(s \times \rho \times t) \times 10^4, \tag{1}$$

where v is the deposition rate (μm/h); W_1 and W_2 are the masses of the plating sample before and after plating, respectively (g); s is the plating surface area (dm^2); ρ is the density of alloy plating layer (g/cm^3); and t is the plating time (h).

Corrosion test was conducted by immersing the coated samples in 5% H_2SO_4, 20% NaOH and 3.5 wt.% NaCl solution for 7 h, respectively. Then, the specimens

were removed from the solutions, cleaned, dried and weighed. The corrosion resistance was determined by the mass-loss and change of surface morphologies.

3. Results and Discussion

Table 1 shows the conditions used in the L9 (3^4) orthogonal test, and some of the evaluation results are also included. The coating (1) has smooth surface, uniform distribution and slightly darker surface. After plating, the solution had a light green color without precipitation. The average coating mass increment is 0.1115 g.

The surface of coating (2) is smooth and evenly distributed, and the reaction is strong during the plating process. The solution after plating is light green and no precipitation occurs. The average coating increment is 0.2974 g.

The surface of coating (3) is smooth and uniform, and a large number of bubbles are formed during the plating process due to the excessive temperature. After plating, the solution is light green with a large amount of white precipitation. The average coating increment is 0.4845 g.

The surface of coating (4) is smooth and even, and a large number of bubbles are formed during the plating process due to the excessive temperature. After plating, the solution is light green with a large amount of white precipitation. The average coating increment is 0.5331 g.

The coating (5) has smooth surface and uniform distribution. After plating, the solution is light green without precipitation. The average coating increment is 0.1476 g.

The coating (6) has a flat surface, uniform distribution, strong reaction in the process of plating, light green solution after plating, and no precipitation. The average coating increment is 0.2845 g.

The surface of the coating (7) is smooth and evenly distributed, and the reaction is strong during the plating process. The solution after plating is light green, and there is no precipitation. The average coating increment is 0.2973 g.

The surface of the coating (8) is smooth and evenly distributed, and a large number of bubbles are formed during the plating process due to excessive temperature.

Table 1. L9 (3^4) orthogonal test.

Factor	$NaH_2PO_2{\cdot}H_2O$ (g/L)	CH_3COONa (g/L)	pH	T (°C)	P (wt.%)	Deposition rate (μm/h)	Hardness (HV)
(1)	30	15	5	60	6.63	11.26	174
(2)	30	20	5.5	70	5.05	22.92	400
(3)	30	25	6	80	6.56	36.19	676
(4)	35	15	5.5	80	7.18	35.41	565
(5)	35	20	6	60	4.55	14.33	195
(6)	35	25	5	70	6.50	20.91	448
(7)	40	15	5.5	80	5.66	20.91	381
(8)	40	20	6	60	8.57	32.78	570
(9)	40	25	5	70	5.25	9.67	178

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Table 2. Corrosion rate (mass loss) of specimens.

Specimen	1	2	3	4	5	6	7	8	9	Matrix
5%H_2SO_4	98.0	45.0	20.7	14.0	174	58.4	60.1	17.3	126.7	2365.5
20%NaOH	0.01	≈0	0.04	≈0	≈0	0.03	0.12	0.02	0.11	2.06
3.5%NaCL	1.84	0.98	1.92	2.45	1.77	1.02	1.26	2.03	1.90	15.68

After plating, the solution is light green with a large amount of white precipitation. The average coating increment is 0.5113 g.

The coating (9) has smooth surface, uniform distribution and slightly darker surface. Light green solution without precipitation. The average coating increment is 0.1357 g.

Table 2 shows the static corrosion rate of the specimen.

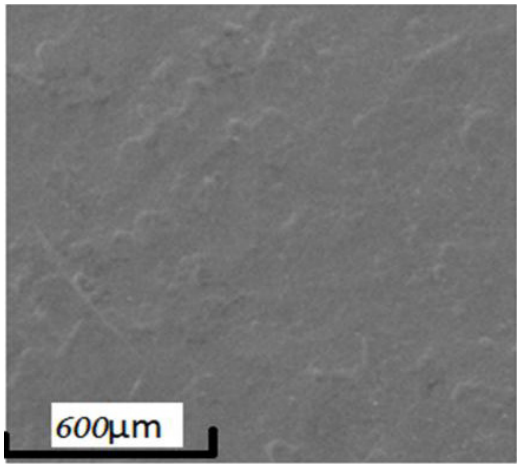


Fig. 1. Morphology of Ni-P coating.

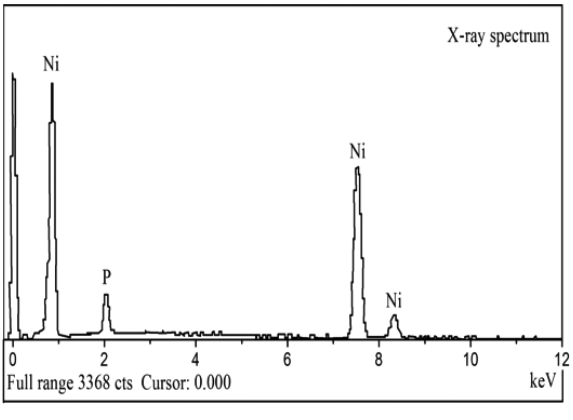


Fig. 2. Energy-dispersive X-ray spectrum of Ni-P coating.

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The corrosion rates of the coatings in 5% H_2SO_4 solution were quite high, and a large number of holes appeared, indication severe corrosion took place. In 3.5% NaCl solution, the coatings were only corroded slightly. In 20% NaOH solution, the coatings were not corroded at all. The results also indicated that the corrosion resistance of low P coatings in neutral salt and alkali solutions is generally higher than that of high P coatings; but the corrosion resistance of low P coatings in acidic solution is not as good as that of high P coatings.

On the basis of the optimized conditions described above, an electroless Ni-P alloy coating with medium temperature and low P content was prepared. The surface morphology and composition of the resulting plating were characterized. Figure 1 shows the Ni-P alloy coating deposited with these conditions has good quality and morphology, with a uniform and smooth surface texture no porosity.

Figure 2 shows the energy-dispersive X-ray spectrum of Ni-P coating deposited by the above process conditions (low P and medium temperature). The microanalysis gives the composition of 5.05 wt.% of P and 94.95 wt.% of Ni.

4. Conclusion

In this work, the preparation process of electroless Ni-P alloy plating at medium temperature and low P was studied by using the single factor experiment and orthogonal experiment. Through constant process parameters debugging, a middle temperature and low P electroless coating were produced, which possesses good hardness, high deposition speed and reasonably good corrosion resistance. The optimized technological parameters include: solution contains 30 g/L NiSO4·$6H_2O$, 30 g/L $NaH_2PO_2{\cdot}H_2O$, 20 g/L CH_3COONa, 15 mL/L $C_3H_5NaO_3$, 15 g/L $(NH_4)_2SO_4$ and 2 mg/L thiourea. The solution has PH = 5.5 and temperature of 70°C.

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